Southwire®

PREPARED FOR NRG | EXECUTIVE WORKING DISCUSSION

PROTECTING TIME TO POWER

A working model for integrated electrical delivery across NRG's generation buildout.

PRESERVE UPSTREAM CERTAINTY THROUGH THE ELECTRICAL DELIVERY CHAIN.

Stephan Hardt | Director, Power Generation Solutions | Southwire

WHY NOW

NRG's growth agenda creates three distinct electrical-delivery environments.

1.5 GW

TEF PORTFOLIO

One project operating; two targeting mid-2028.

5.4 GW

NEWBUILD DEVELOPMENT

Turbine and EPC access supports a runway through 2032.

25.8 GW

OPERATING FLEET

Approximately 1-2 GW of upgrades under evaluation.

DIFFERENT ASSET STATES. ONE REQUIREMENT.

Preserve schedule certainty through energization.

Specifications, capacity, releases, reels, package interfaces and field readiness still have to converge on COD.

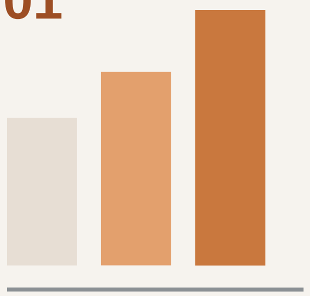
Source: NRG 2Q26, Aug. 4, 2026

Southwire POWER GENERATION SOLUTIONS

THREE RISKS

Three execution risks can erode certainty between design and energization.

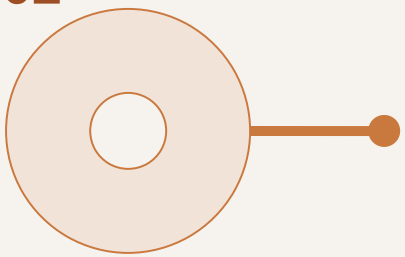
01



LABOR PRESSURE

Long pulls, dense terminations and compressed turnover windows concentrate labor when flexibility is lowest.

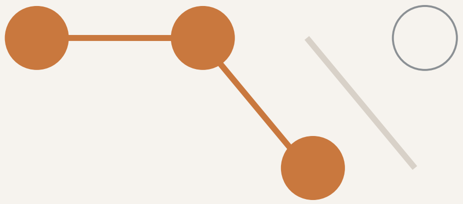
02



MATERIAL + INSTALLATION FIT

Ratings, routes, reel lengths, accessories and releases must match the workface—not only the BOM.

03



FRAGMENTED COORDINATION

Civil, electrical, OEM and contractor decisions can converge after routing and procurement choices are locked.

LATE HANDOFFS BECOME SCHEDULE AND COMMISSIONING EXPOSURE.

Labor availability | material readiness | decision timing

TIME TO POWER IS AN ELECTRICAL EXECUTION PROBLEM BEFORE IT IS A CABLE-BUYING PROBLEM.

- Interfaces cross packages, zones, contractors and turnover paths.
- Late material decisions return as handling, rework and commissioning exposure.
- Managing the workstream as a system can protect schedule better than isolated purchase-order optimization.

DESIGN — MATERIAL — WORKFACE — ENERGIZATION

PLANT-ZONE COVERAGE

Six clusters map a routed 3×1 electrical reference model.

1, 16

GRID + SITE BACKBONE

Switchyard, GSU/grid tie and MV corridors
HV/EHV · MV · protection/control · fiber · grounding

2, 3, 10

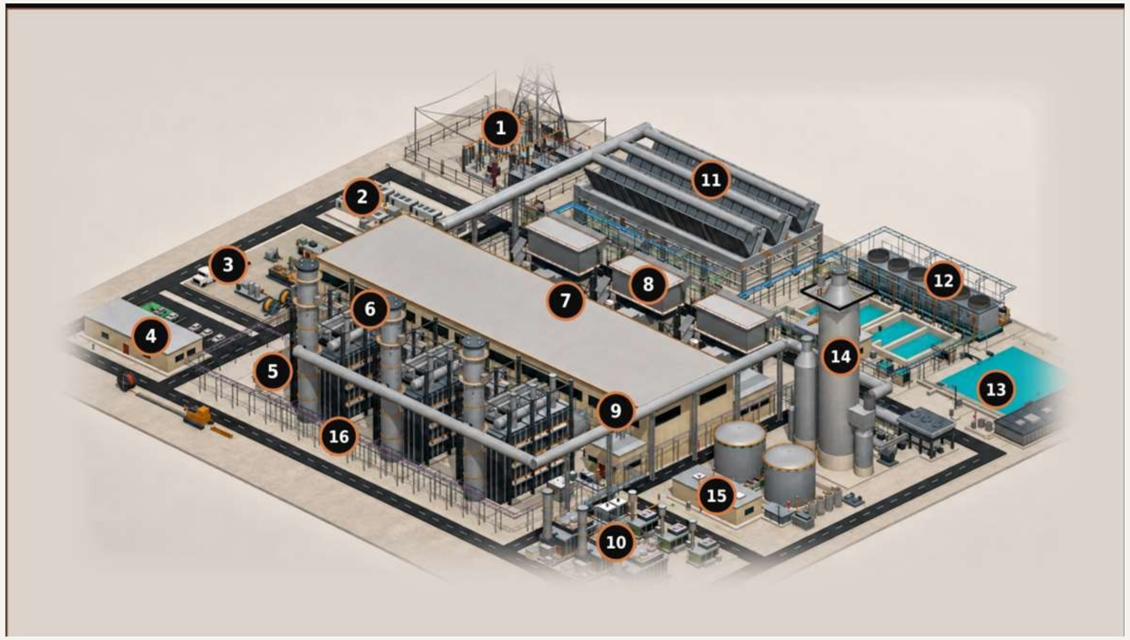
MODULAR + LOGISTICS

BESS, reel staging, prefab/skids and modular power
DC/MV/LV · controls · fiber · engineered sets

4, 5, 9

BUILDINGS + DISTRIBUTION

Control building, e-house and MCC/VFD/UPS
MV/LV/VFD · essential AC/DC · life safety · BAS/data



6, 7, 8

GENERATION ISLAND

HRSG, turbine hall and GT inlet
MV/LV/VFD · I&C · F&G/ESD · CEMS · heat trace

11, 12, 13

HEAT REJECTION + WATER

ACC, cooling and water/wastewater
Repeated fan/pump power · VFD · controls · wet runs

14, 15

OPTIONAL + AUXILIARY

Carbon capture and gas metering
Classified power/control · instrumentation · F&G · fiber

CABLE CLASSES

HV/EHV | MV 5-35 kV | LV 600 V | VFD | essential AC/DC | I&C | fiber/network | life safety | grounding

Illustrative—not NRG/IFC. Excludes generator-output bus, OEM wiring and contractor methods.

ONE COORDINATED FLOW

A coordinated electrical delivery system aligns decisions from planning through turnover.

Southwire connects application support, material planning, sequenced logistics and field readiness—without changing project accountability.

1. **PLAN + FORECAST** — Scope and capacity visibility
2. **APPLICATION ALIGNMENT** — Codes, duty and alternates
3. **MATERIAL + REEL STRATEGY** — Routes, cuts and reel plan
4. **RELEASE + LOGISTICS** — Sequence by zone and milestone
5. **INSTALLATION READINESS** — Labels, kits and pull planning
6. **FIELD SUPPORT + TURNOVER** — Feedback, testing and records

SOUTHWIRE COORDINATION RAIL

Shared data | governed handoffs | field feedback

NRG governance, EPC/EOR/OEM design authority and contractor means and methods remain intact.

FIELD-ENGINEER WHAT IS UNIQUE. PREFABRICATE WHAT REPEATS.

THE OUTCOME

The value to NRG is fewer execution surprises—not more supplier services.

Three owner lenses matter: schedule exposure, installed effort and lifecycle continuity.

ALIGN APPLICATIONS + INTERFACES EARLY →	Fewer late changes and handoff failures
GATE CAPACITY + RELEASES →	More predictable material availability and sequencing
PLAN REELS + REPEATABLE SETS TO THE WORKFACE →	Less handling and avoidable rework
CARRY FIELD RECORDS THROUGH TURNOVER →	Cleaner commissioning and lifecycle continuity

OUTCOME HYPOTHESES TO VALIDATE ON ONE LIVE NRG SCOPE.

OWNER ECONOMICS

Cable is a small cost category with outsized execution consequences.

The owner case is not unit price alone; it is how electrical decisions influence COD readiness, installed effort, capital certainty and lifecycle continuity.

01 COD / REVENUE START

Capacity visibility, release timing and installation readiness can influence energization milestones.

02 INSTALLED COST

Routes, reels, cuts, pulls, congestion and rework often matter more than purchase price alone.

03 CAPITAL CERTAINTY

Earlier scope visibility supports commodity planning, material allocation and contingency discipline.

04 AVAILABILITY / LIFECYCLE

Turnover records, spares, condition screens and replacement planning support long-term continuity.

VALIDATE THE VALUE ON ONE LIVE NRG SCOPE — DO NOT ASSUME SAVINGS.

ILLUSTRATIVE ADJACENT-MARKET BENCHMARK | NOT AN NRG RESULT

Adjacent experience indicates where to test—not what NRG will save.

Field hours per electrical spine in one repetitive modular-generation scope

4,400

OBSERVED FIELD HOURS
SITE-BUILT BASELINE

~880

MODELED FIELD HOURS
WITH FACTORY-BUILT ASSEMBLY

WHAT IT SUGGESTS

Stable, repeatable interfaces may be candidates for offsite work.

WHAT IT DOES NOT PROVE

Net NRG savings, schedule reduction or critical-path removal.

Rebuild the case with NRG scope, labor rates, schedule logic and acceptance criteria.

ONE LIVE SCOPE

One live scope can prove—or disprove—the case before NRG expands it.

NRG selects the starting scope: one live package, conversion/uprate screen or planned-outage replacement.

01	02	03	04
SELECT SCOPE	**WALK THE WORK**	**MAP THE RISK**	**DECIDE**
One active electrical package	People closest to execution	Interfaces, releases, constraints + repetition	Stop, standardize or pilot

WRITTEN READ-BACK

- **Milestone exposure**
- **Route, reel + release constraints**
- **Repeatable-interface candidates**
- **Decision gaps + recommendation**

STOP
STANDARDIZE
PILOT

No presumption that the answer is prefab—or that Southwire belongs in the solution.

THE ASK

ONE SITE WALK.
ONE HOUR.
NO SALES PRESENTATION.

Choose one live electrical scope and bring the people closest to execution.

INPUT	**One live scope**
OUTPUT	**Written execution-risk read-back**
DECISION	**Stop, standardize or define a bounded pilot**

NO PRICING EXERCISE. NO BROAD COMMERCIAL COMMITMENT.

Stephan Hardt | Director, Power Generation Solutions | Southwire

stephan.hardt@southwire.com | +1 470-439-8488

Southwire POWER GENERATION SOLUTIONS

SOUTHWIRE AT A GLANCE · PUBLISHED FIGURES · 2025 REVENUE PER FORBES 2026

A family-held American manufacturer, founded in 1950.

Roy Richards founded Southwire after the poles his crews built stood wireless for months after World War II. Starting with 12 employees and three secondhand machines in Carrollton, Georgia, Southwire remains family-held and vertically integrated — from copper rod through finished cable.

1950	9,000+	$9.7B	12	7
FOUNDED · CARROLLTON, GA	EMPLOYEES WORLDWIDE	2025 REVENUE · FORBES EST.	INDUSTRIES SERVED	COPPER MARK SITES

WHAT WE MAKE

High voltage underground — 69–500 kV cable, accessories, installation engineering and field testing.

Medium & low voltage — 5–35 kV EPR/XLPE; 600 V power and control; tray-rated TC-ER.

Instrumentation & comms — shielded pairs and triads, thermocouple, industrial Ethernet and fiber.

Overhead conductors — C7 composite-core, ACSS/TW, vibration-resistant, high-emissivity.

Classified & circuit integrity — MC-HL, two-hour-rated cable and splice kits.

Modular & factory-built power — fine-strand flexible, VFD and tray cable; whips and job kits.

Data center assemblies — UL-listed, built, labeled, tested, drum-packed to schedule.

Accessories & raceway — lugs, grounding, duct-bank accessories and pulling gear.

WHAT WE DO

- HV underground transmission
- Cable rejuvenation
- Contractor solutions
- Speed to Power services
- Project services
- Storm activation team
- Field assessment services
- Reel tracking system
- CableTechSupport™
- Solutions University
- Utility services
- Grid resiliency assessments

INDEPENDENT COVERAGE — FORBES, JUL 5, 2026: *“AI Data Centers Are Set To Electrify This $13 Billion Family’s 76-Year-Old Company”* READ ON FORBES ↗

COPPER TECHNOLOGY	U.S. POWER GRID	AMERICAN HOMES
Half of the world's copper rod passes through a Southwire SCR® system.	**We produce half of the wire and cable that moves U.S. electricity.**	**Half of American homes contain wiring produced by Southwire.**

SOURCE DISCIPLINE

NRG context is current, attributed and non-additive.

1.5 GW TEF PORTFOLIO	One operating project and two projects targeting mid-2028. Do not describe the entire portfolio as under construction.
5.4 GW NEWBUILD DEVELOPMENT	Turbine and EPC access supports a runway through 2032. Do not call all 5.4 GW committed or under construction.
25.8 GW OPERATING FLEET	NRG's reported operating-generation fleet after the LS Power acquisition.
~1-2 GW UPGRADE POTENTIAL	CT-to-CCGT conversions and traditional uprates are under evaluation—not a committed pipeline.

DO NOT SUM THESE FIGURES: THEY REPRESENT DIFFERENT ASSET STATES AND OVERLAP CONCEPTUALLY.

DECISION RIGHTS

Clear decision rights preserve existing project accountability.

NRG	Owner priorities, portfolio standards, risk tolerances, acceptance criteria and investment decisions.
EPC / EOR	Design authority, calculations, specifications, routing basis and approved technical design.
OEM	Equipment-package interfaces, internal wiring, approved terminations and warranty boundaries.
ELECTRICAL CONTRACTOR	Installation means and methods, pull planning, workface execution, testing support and field feedback.
SOUTHWIRE	Application support, material and capacity visibility, engineered reel strategy, sequenced delivery and selected field support.

Southwire connects selected decisions; it does not replace owner, EPC, OEM or contractor accountability.

DEFINED SUPPORT

Southwire can support selected delivery stages without taking design authority.

ALIGN

Application and BOM review

Alternative-construction evaluation

Multidisciplinary cable support

COORDINATE

Project cable management

Inventory and release coordination

Reel, cut and sequence planning

PREPARE + SUPPORT

Cut-to-length, striping and labeling

Kitting by package or work sequence

Selected cable-in-conduit, pulling + field support

PROCUREMENT + SUSTAINABILITY SUPPORT

Product-level EPDs and responsible-production evidence can support procurement where applicable to the selected product and manufacturing site.

Capability does not equal outcome. Availability, geography, contractual scope and technical fit must be confirmed for each project.

OBSERVED, MODELED, VALIDATED

Evidence is credible only when its boundaries stay visible.

OBSERVED + PUBLISHED

- Name the scope, source, permission and unit of measure.
- Separate measured performance from arithmetic and inference.
- Keep customer- and line-specific conditions attached to each result.

MODELED + DIRECTIONAL

- State every input, exclusion and scope-equivalence assumption.
- Include engineering, factory work, freight, FAT and residual field work.
- Test whether the activity leaves the critical path before treating elapsed-time compression as schedule impact.

DO NOT TURN A SCREENING MODEL INTO AN NRG OUTCOME.

Release gate: source permission, scope equivalence and NRG-specific validation approved.

The plant-zone map is grounded in a routed 3×1 reference model.

Every figure below comes from the routed Rev 89 geometry — built before any RFQ existed.

96 SCHEDULE RECORDS	NGCC master cable schedule Rev M21 — geometry-governed quantities, not factored estimates.
1,784 CIRCUITS ROUTED	2,478 physical runs routed through the Rev 89 site geometry, across every plant zone in slide 5.
1,136,255 PROCUREMENT FT	Approximately 1.14M ft of cable derived from routed lengths plus engineering allowances.
563,091 LB COPPER	About 0.50 lb of copper per foot on average — commodity exposure quantified before procurement.

Published Southwire record: 15M+ ft of HV underground cable with zero in-service failures.

THE MODEL BEHIND SLIDE 5 — CONCEPT, NOT FOR CONSTRUCTION; STOCK NUMBERS REQUIRE CONDUCTOR SIZING.

PLANT ZONES & KEY CABLE APPLICATIONS

WHAT EACH ZONE POWERS — AND WHY THE CABLE CHOICE MATTERS.

APPENDIX A

SK-120

3x1 NGCC · REV 89 SITE MODEL · CONCEPT, NOT FOR CONSTRUCTION

1 SWITCHYARD, GSU BANK & GRID TIE

Overhead gen-tie conductors or HV/EHV underground cable, plus protection/control, fiber and grounding. POI boundaries, accessories and testing can gate energization.

ACSR/ACSS • HV/EHV XLPE • TC-ER control • fiber • bare copper

2 BESS YARD

Flexible BESS DC cable, MV collection, LV auxiliaries, controls and fiber. Repeated containers and PCS skids support a standardized DC/LV/control/fiber spine; MV collection follows the plant one-line.

RenewaFLEX 2 kV • DLO • MV-105 • XHHW-2/TC-ER • fiber

3 REEL STAGING, LAYDOWN & PREFABS / SKIDS

Zone-tagged reels, engineered cut lengths, spine sections and prefab / skid cable sets span all cable classes. Installation-sequenced staging reduces splices, scrap, handling and congestion.

MV-105 • XHHW-2 • TC-ER • VFD cable • MC-HL

4 ADMIN BUILDING & CONTROL ROOM

Low-voltage fire alarm, access control, HVAC/BAS and data join building power here. Separation protects life-safety and controls.

THHN/THWN-2 • Genesis fire alarm • Genesis access control • Genesis HVAC/BAS • fiber/data

5 E-HOUSE — MODULAR ELECTRICAL BUILDING

MV feeders, LV/VFD power, controls, fiber, station DC and panel wire. Defined E-house interfaces make it the spine head-end, with pre-engineered cable sets reducing field terminations.

MV-105 • XHHW-2/TC-ER • shielded VFD • PLTC/ITC • SIS

6 HRSG TRAINS & STACKS

Motor/valve power, lighting/small power, heat trace, thermocouple/RTD, CEMS and F&G/ESD cable serve HRSG skids. Pre-engineered harnesses simplify interfaces; heat and vertical runs still govern supports.

XHHW-2/TC-ER • MC-HL • PLTC/ITC F&G/ESD • thermocouple/RTD • heat-trace

7 TURBINE HALL — GAS & STEAM TURBINES

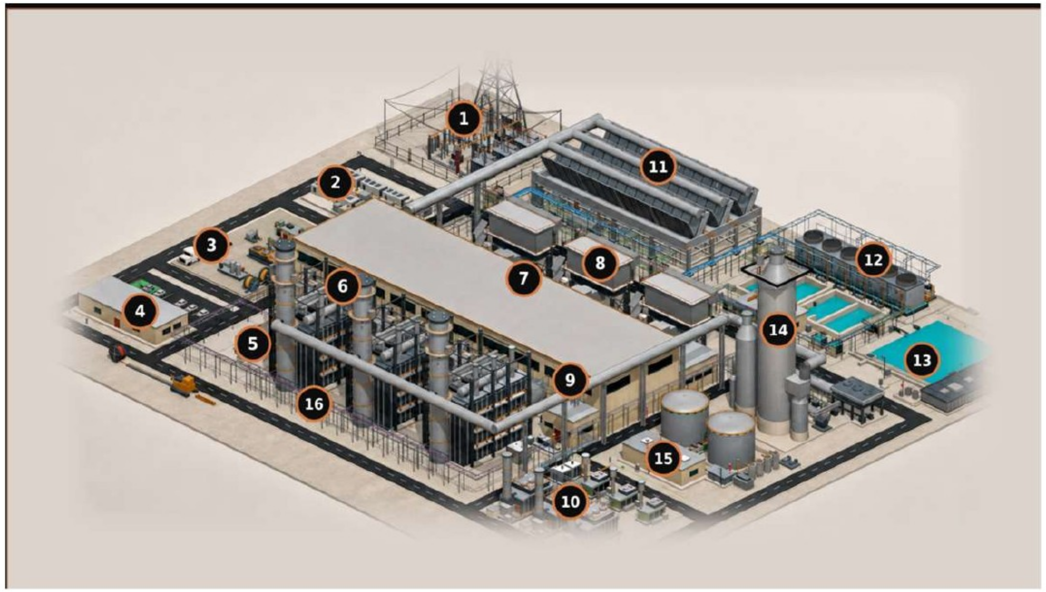
MV/LV/VFD power, generator/turbine controls, F&G/ESD, lighting/small power and festoon cable serve turbine-hall skids. A skid spine simplifies auxiliaries; generator output remains isolated-phase bus.

MV-105 • ARMOR-X MV VFD • XHHW-2/TC-ER • PLTC/ITC F&G/ESD • festoon

8 GT AIR INLET SECTION

Fan and heater power, controls, plus shielded pressure and temperature instrumentation. Repeated inlet houses support standardized drop assemblies tied to a common control interface.

TC-ER • shielded VFD • PLTC/ITC • thermocouple/RTD • fiber



CABLE CLASSES CARRIED BY THE SCHEDULE

Bus duct and the overhead getaway are boundary only — excluded from all totals

- MV 5–35 kV
- LV 600 V
- I&C / instrument
- DC 125 V
- Essential AC
- Fibre / network
- Grounding, bare

HV UNDERGROUND 69–500 kV • MV EPR 133% 5 / 15 / 25 / 35 kV • DUCT BANK, DIRECT BURIED AND TRAY

Published Southwire record: 15M+ ft HV underground, zero in-service failures

96	1,784	2,478	1,136,255	563,091	~0.50
SCHEDULE RECORDS	CIRCUITS ROUTED	PHYSICAL RUNS	PROCUREMENT FT	LB COPPER	LB CU / FT AVG

SCHEDULE BASIS

QUANTITIES FROM THE NGCC MASTER CABLE SCHEDULE REV M21, GEOMETRY-GOVERNED. SPECIFICATION NUMBERS ARE SOUTHWIRE FAMILY REFERENCES — A STOCK NUMBER REQUIRES CONDUCTOR SIZING AND IS DELIBERATELY NOT SHOWN. ZONE NUMBERS ARE KEYED TO THIS VIEW.

9 MCC, MOTOR VFD & UPS ROOM

LV/MV motor feeders, VFD cable, essential AC/DC and fire-alarm/control CI wiring originate here where required. Outgoing sets simplify pulls; shielding and grounding still manage PWM stress and EMI.

MV-105 VFD • ARMOR-X VFD • Machine Flex VFD • DLO 2 kV • Circuit Defender

10 MODULAR POWER YARD — GENSETS, SIMPLE CYCLE, BTM, BLACK START, FUEL CELLS

MV/LV, control, fiber and grounding can be coordinated into prefabricated electrical spines for modular packages and skids. Consolidated terminations reduce field pulls and speed installation.

MV-105 • XHHW-2/TC-ER • DLO/RHH/RHW-2 • PLTC/ITC • fiber

11 ACC — AIR-COOLED CONDENSER

VFD power, controls, vibration/temperature instrumentation and fiber serve repeated fan cells. One trunk-and-drop electrical spine can repeat bay by bay across the ACC.

shielded VFD • XHHW-2/TC-ER • PLTC/ITC • vibration/RTD • fiber

12 CHILLERS & COOLING TOWERS

Chiller/pump/tower feeders, MV/LV VFD cable, HVAC/BAS controls and submersible cable serve packaged skids and cells. Modular spines speed installation; wet runs remain field-engineered.

MV-105 • ARMOR-X MV VFD • XHHW-2/TC-ER • PLTC/ITC HVAC/BAS • submersible pump

13 WATER TREATMENT, PROCESS TANKS & WASTEWATER POND

Pump/VFD power, TC-ER cable, MC-HL where classified, shielded instrumentation and submersible cable. Dosing and pump skids can use prewired power/control/instrument whips; wet-pit and submersible runs stay field-engineered.

XHHW-2/TC-ER • shielded VFD • MC-HL • PLTC/ITC • submersible

14 CCS — ABSORBER / REGENERATOR

LV/MV feeders, VFD cable, MC-HL, instrumentation, F&G/ESD, heat trace and fiber serve CCS packages. If OEM modules and skid interfaces are frozen, they can form a repeatable CCS spine; reserve corridor and tray capacity early.

MV-105 • ARMOR-X MV VFD • MC-HL • PLTC/ITC F&G/ESD • heat-trace

15 GAS METERING / METER HOUSE

One umbilical combines metering-skid power, F&G/ESD and instrumentation. Heat trace and sealing protect accuracy.

MC-HL power/control • ARMOR-X instrumentation • PLTC/ITC F&G/ESD • heat-trace • fiber

16 SITE CABLE CORRIDOR & MV DISTRIBUTION

MV backbone, LV auxiliaries, lighting/small power, controls, fiber and grounding cross site corridors. Route geometry, thermal design and pull limits shape cable and reel strategy.

MV-105 • XHHW-2 lighting/small power • TC-ER control • single-mode fiber • bare copper

FASTER TIME TO POWER. LOWER EXECUTION RISK.
ENGINEERED FOR LONG-TERM RELIABILITY.

stephan.hardt@southwire.com | +1 470-439-8488
Stephan Hardt - Power Generation Solutions

SK-120 · 3×1 NGCC · Rev 89 site model · 16 zones with per-zone cable applications · concept, not for construction · print at 34" for site-walk use.

INTERNAL ILLUSTRATIVE MODEL | NOT A CUSTOMER OR NRG RESULT

Repeatable skid interfaces are a prefabrication candidate—not a promised result.

Supplied 50 MW power-distribution skid scenario | timing is directional and unvalidated

8–12 wks

SUPPLIED FIELD-BUILD MODEL
INSTALL + COMMISSION

→

2–4 wks

SUPPLIED PREFAB MODEL
INSTALL + COMMISSION

WHAT IT SUGGESTS

Pretested, prelabeled harnesses may reduce onsite interfaces and congestion.

WHAT IT DOES NOT PROVE

Net savings, a six-to-eight-week gain or lower commissioning risk.

Validate scope, crew plan, factory cost, freight, work rules, commissioning and schedule logic.

PUBLISHED ADJACENT CASE | ANONYMOUS UTILITY | NOT AN NRG RESULT

A 138-kV crossing case shows where a conductor screen can change the rebuild question.

Published Southwire river-crossing comparison under specific engineering assumptions

730 A

EXISTING 795 KCMIL DRAKE ACSR
98.0 FT CALCULATED FINAL SAG

→

1,050 A

477 KCMIL C7 SOLUTION
33.5 FT CALCULATED FINAL SAG

WHAT IT SUGGESTS

Advanced conductor may avoid a full rebuild where line conditions permit.

WHAT IT DOES NOT PROVE

NRG transfer capability, corridor fit, cost or schedule.

WHY AN IPP CARES: GENERATOR-FUNDED NETWORK UPGRADES · THE GEN-TIE THE GENERATOR OWNS · UPRATE THERMAL LIMITS · IN-SERVICE CORRIDOR HEADROOM VS. A NEW QUEUE POSITION.

Require transmission-owner power-flow, structure, clearance, terminal, outage and permit validation.

ONE SYSTEM · THREE ASSET STATES

The governance model stays constant; the deployment path changes by asset state.

NEW BUILD	Standardize package interfaces, reserve capacity and sequence releases by workface.
CONVERSION / UPRATE	Revalidate duty, routes and terminals; map brownfield interfaces; stage work around outage constraints.
OPERATING FLEET / OUTAGE	Screen condition and criticality; evaluate replacement, eligible rejuvenation or outage spares; align material to outage windows.
COMMON GOVERNANCE RAIL	Owner priorities, EPC/EOR design authority, OEM interfaces and contractor means and methods remain intact.

SAME COORDINATION RAIL. DIFFERENT ENGINEERING BASIS, RISK GATES AND OWNERS.

CEDAR BAYOU UNIT 5 · BAYTOWN, TEXAS (CHAMBERS COUNTY)

Cedar Bayou Unit 5 is one potential live-scope candidate.

A 721 MW combined-cycle brownfield expansion at NRG's existing Cedar Bayou station — the anchor TEF unit in the NRG / GE Vernova / Kiewit framework, in active construction toward a mid-2028 COD.

PROJECT FACTS — VALIDATED AUG 2026

721 MW CCGT · BROWNFIELD

GE Vernova H-class train at the existing station; ERCOT Houston Load Zone; existing 345 kV switchyard, cooling water and site infrastructure.

$562M TEF LOAN · 3% · 20-YEAR

Signed September 2025 — 60% of the $936M project cost; part of $1.15B TEF financing plus $66M tax abatements across the three-unit program.

COD TARGET MID-2028

First TEF unit to advance construction; T.H. Wharton (415 MW) completes mid-2026, Greens Bayou 6 follows in 2028.

EPC: TIC / KIEWIT · OEM: GE VERNOVA

Delivered under the 5.4 GW development framework — 1.2 GW in service by 2029, 1.2 GW by 2030, 3 GW more through 2032.

SCOPE THAT WOULD BE IN PLAY

- HV generator leads and brownfield 345 kV switchyard tie-in for the H-class train
- MV plant auxiliary distribution, 5–35 kV, across the new unit and interfaces
- LV 600 V power and control — turbine hall, HRSG and balance of plant
- I&C, DCS and fiber — integration with the existing station and ERCOT SCADA
- Grounding-grid extension into the heritage plant grid
- Sequenced reels, cut lengths and kits against 2026–2028 procurement windows
- Post-COD continuity — records, spares and MRO carried into the operating fleet

Validated against public sources, Aug 2026: PUCT / Texas Energy Fund announcements, Utility Dive, NRG investor materials. Capacity reported 689–721 MW across sources; 721 MW per the PUCT loan announcement.

ILLUSTRATIVE CANDIDATE ONLY — NRG SELECTS THE STARTING SCOPE.

EARLY ENGAGEMENT, UNDERWRITING-GRADE DOCUMENTATION

What early engagement includes — and what finance gets out of it.

Technical readiness converts into underwriting inputs: cost, schedule, sourcing and carbon visibility before the EPC bid hardens contingency.

PRE-EPC OFFERINGS

- Master planning workshop — conceptual design and interface mapping for complex sites
- Conceptual cable solution design and electrical spine architecture
- Procurement risk register and schedule visibility
- Capacity allocation and commodity (Cu/Al) exposure hedging support
- Domestic supply strategy — U.S.-made resilience for TEF and local-content goals

DOCUMENTATION FOR FINANCE + ESG

- Transparent BOMs and cost-allocation records supporting asset classification and depreciation — tax treatment confirmed per project
- Verified EPDs for low-embodied-carbon procurement
- Scope 3 cradle-to-gate carbon data for ESG-linked financing and hyperscaler offtaker requirements
- Recycled-content integration and optimized logistics (circular economy)
- Auditable domestic-content documentation for TEF and investor reporting

Conceptual commercial positioning — offerings, documentation and tax applicability are scoped and confirmed per project.

COST, SCHEDULE, SOURCING AND CARBON VISIBILITY — BEFORE THE EPC BID PRICES THE UNCERTAINTY AS CONTINGENCY.

ONE PLATFORM — PLANT SCOPE AND PACKAGED SCOPE

One platform from generator to grid — and inside every package.

Every run in the plant-zone map resolves to a named product family — one supplier, one spec basis, domestic manufacture.

PLANT SCOPE — GENERATOR TO GRID

HVUD HIGH-VOLTAGE UNDERGROUND · 69–400 kV

XLPE underground interconnect; HVUD3 (138 kV) and HVUD4 (230 kV) carry gen-tie, behind-the-meter and bulk interconnect duty.

MV-105 · NL-EPR · 5–35 kV

The plant collection and auxiliary backbone — 105°C rated, low-friction jacket, tray and duct-bank installation economics.

LV 600 V POWER, CONTROL & BUILDING WIRE

Feeders, motor circuits and unit auxiliary distribution across the plant.

I&C, COMMS & TRANSMISSION CONDUCTOR

DCS, protection and metering cable; industrial Ethernet and fiber; ACSR / ACSS / OPGW where overhead is the route.

GROUNDING & BONDING

Bare copper and bonding conductor — the grid that ties the whole electrical system to earth.

PACKAGED & MODULAR SCOPE — 8 APPLICATION FAMILIES

2.4–35 kV AND 600 V–2 kV · BUY AMERICA

Every wire and cable application that lives inside — or connects — factory-built power packages, kitted to build sequence.

E-HOUSES · CONTAINERIZED SUBSTATIONS · SWITCHGEAR

Power control rooms, MV switchgear lineups, LV switchboard and MCC packages.

TRANSFORMER · BESS · GENSET · PROCESS SKIDS

Transformer packages, BESS containers, generator packages and process / pump skids.

SIMPULL, PRE-CUTS & KITTING

Low-friction pulling technology, cut-to-length labeled reels and staged kits — energization-ready, less field labor.

ONE SUPPLIER, WHOLE PACKAGE

MV, LV, control, VFD, grounding, data and DC from a single domestic source — one procurement line item per package.

TWO LEAVE-BEHIND CATALOGUES ACCOMPANY THIS DECK — IPP CABLE CATALOGUE (PLANT) · PACKAGED POWER E-CATALOGUE (SKIDS).

REJUVENATION, BROWNFIELD & SERVICE — FOR THE 25.8 GW FLEET

Extend asset life without a greenfield rebuild.

Aging cable systems, upgrades and modular additions across an operating fleet — planned around outage schedules, while the plant stays available.

ELECTRICAL SPINE REJUVENATION

Assessment, planning and phased replacement of aging MV / LV cable, grounding, tray and fiber — while the plant remains operational.

CABLE REPLACEMENT PROGRAMS

Fast-track domestic supply of power, control and instrumentation cable for planned outages or emergency replacement.

BROWNFIELD UPGRADES & REPOWER

Electrical scope for turbine upgrades, generator replacements, emissions retrofits and balance-of-plant modernization.

MODULAR POWER ADDITIONS

E-houses, skids, switchgear, UPS and black-start capability — auxiliary loads or new behind-the-meter offtakers.

RELIABILITY & CONDITION SUPPORT

System assessments, thermal-imaging coordination, failure analysis and prioritized replacement planning.

SERVICE & LONG-TERM AGREEMENTS

Support contracts, inventory programs and rapid-response replacement capability for operating fleets.

Engagements begin with one focused 60–90 minute session across the fleet, then tailored proposals per plant — phased to outage schedules.

MINIMIZED DOWNTIME · EXTENDED ASSET LIFE · LOWER TOTAL COST THAN FULL REPLACEMENT — PHASED TO OUTAGE SCHEDULES.

GROSS FIELD-LABOR MATH — PER CENTRAL SPINE

What the 4,400 → 880 hours are worth, before net.

The arithmetic behind slide 9, at a $95/hr fully burdened field-labor rate. This is the template A5 requires to be rebuilt with NRG inputs — not a result.

TODAY · FIELD-BUILT

$418,000

field labor per spine

4,400 man-hours × $95/hr

27 installers × 2 shifts × 2 weeks

High variability, rework risk

WITH CONNECTORIZED ASSEMBLY

$83,600

field labor per spine

~880 man-hours × $95/hr

~6 installers × 1 shift × ~3 days

Factory-validated, plug-and-play

GROSS DELTA PER SPINE

$334K

~80% of field-labor cost

IF IT REPEATS (GROSS, ILLUSTRATIVE)

10 spines → $3.3M
50 spines → $16.7M
100 spines → $33.4M

ASSUMPTIONS · $95/hr fully burdened (wage + benefits + overtime + supervision) · 4,400 hours observed on one site-built spine in an adjacent market, May 2026 · ~80% field-labor reduction typical for prefabricated interconnect · one spine = one central electrical spine serving repeated generation modules.

NOT NET. Subtract the assembly premium, engineering, factory labor, freight, factory acceptance and commissioning — and prefabrication does not automatically remove an activity from the critical path.

GROSS TEMPLATE ONLY — REBUILD WITH NRG SCOPE, LABOR RATES AND SCHEDULE LOGIC BEFORE ANY CLAIM (SEE A5).

DOCUMENTS AND REFERENCES BEHIND THIS DECK

Everything referenced here exists as a document you can have.

LEAVE-BEHIND DOCUMENTS — AVAILABLE ON REQUEST

IPP Cable Catalogue — the plant platform: HVUD high-voltage underground classes 69–400 kV, MV-105 NL-EPR, balance of system, and the pre-FEED engagement model. Behind A13.

Packaged Power e-Catalogue — wire and cable for skids, e-houses, containerized substations, BESS and gensets; SIMpull, pre-cuts and kitting. Behind A13 and the skid scenario (A8).

Rejuvenation, Brownfield & Service Offerings for IPP Portfolios — the operating-fleet program: spine rejuvenation, cable replacement, repower support, condition programs. Behind A14.

Integrated Power Infrastructure e-Catalog — pre-EPC master planning workshop, spine architecture, procurement risk register, finance and ESG documentation. Behind A12.

SK-120 · Plant Zones & Key Cable Applications — the 34-inch site-walk poster: 16 zones, per-zone cable applications, routed 3×1 basis. Reproduced at A7.

Cedar Bayou Unit 5 project reference — public-record project facts and scope selections, validated Aug 2026. Behind A11.

PUBLIC REFERENCES

southwire.com/services/project-services · Project services

southwire.com/services · Services & technical support

southwire.com/sustainability/product-transparency · EPDs & product transparency

southwire.com/sustainability/doing-right · Responsible production

INDEPENDENT COVERAGE · FORBES · JUL 5, 2026

"AI Data Centers Are Set To Electrify This $13 Billion Family's 76-Year-Old Company"

READ ON FORBES ↗

Illustration: Macy Sinreich for Forbes; images: Getty Images.

Fact base: NRG 2Q26 investor materials (Aug 4, 2026), PUCT / Texas Energy Fund announcements, Utility Dive coverage — as constrained by A2 and A5.

EVERY CLAIM IN THIS DECK TRACES TO ONE OF THESE DOCUMENTS — ASK AND WE WILL SEND THE SOURCE.